ASME V&V 40 · NASA-STD-7009B · OPEN SOURCE · APACHE-2.0

Unit of *Assurance* (UofA)

An open-source way to package simulation credibility evidence so a reviewer can verify it *without reading 200 pages of prose.*

Vishnu Vettrivel · Doctoral Candidate, Systems Engineering, George Washington University

THE PROBLEM

Reviewers don't reject your simulation. They reject your evidence package. Standards tell you *what* to assess. Your tools capture *what ran*. Neither captures *why you believe it*, and that's where submissions stall.

C1 · PACKAGES

Decisions as artifacts

Your credibility evidence becomes a portable, signed object you hand a reviewer. Ed25519 + SHA-256. Tool-independent. Tamper-evident.

C2 · DETECTS GAPS

A rule engine for weakeners

Forward-chaining rules catch missing UQ and unsupported claims. Surfaces compound risks no standalone SPARQL query can find.

C3 · COMPARES

Two contexts of use, side by side

When the bar moves, see exactly which factor moved with it. Diff one COU against another in 30 seconds.

```
uofa rules morrison/cou2

$ uofa rules morrison/cou2.jsonld

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  Weakener Detection Report
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SUMMARY: 18 weakeners detected
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  Critical:  9
  High:      7
  Medium:    2

⚡ W-PROV-01    [Crit]    7 hits
⚡ COMPOUND-01  [Crit]    2 hits
⚠ W-EP-04      [High]    6 hits
⚠ W-ON-02      [High]    1 hit
⚠ W-AL-02      [Med]     1 hit
⚠ W-CON-04     [Med]     1 hit

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⚡ 2 compound inferences — chained,
  not detectable by standalone SPARQL.
```

WHY IT MIGHT FIT YOU

Works in medical device (ASME V&V 40) and aerospace (NASA-STD-7009B) today. Runs locally, so evidence never leaves your environment. Sits alongside your PLM and SPDM — it doesn't replace them.

ZERO INSTALL	ON YOUR MACHINE	WORKED EXAMPLE
codespaces.new/ cloudronin/uofa	pip install uofa uofa setup	uofa.net/demo

SCAN FOR DEMO

The ask: Run it on one of your own models — even a small one. Email me what breaks or what's missing.

Vishnu Vettrivel
support@uofa.net
uofa.net · github.com/cloudronin/uofa